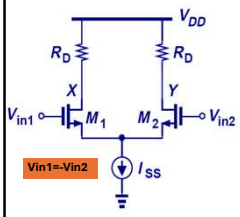
Differential amplifier

MOS Differential Pair:



- The sources of M1 and M2 are tied to a constant current source rather than to ground.
- We call I_{SS} the "tail current source."
- The sum of the transistor currents is equal to the tail current.
- For each differential pair, we begin with a qualitative analysis and subsequently formulate the large-signal and small-signal behavior.
- We also assume each circuit is perfectly symmetric, i.e., the transistors are identical and so are the resistors.

1

Differential amplifier

Qualitative analysis:

Operation for common mode input signal:

- Initially assume perfect symmetry.
- **MOS pair with the two inputs tied to V_{CM}, yields**

$I_{D1} + I_{D2} = I_{SS}$

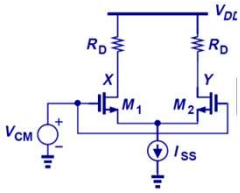
$I_{D1} = I_{D2} = I_{SS}/2$

$V_X = V_{DD} - I_{D1}R_D$

$V_Y = V_{DD} - I_{D2}R_D$

$V_X = V_Y = V_{DD} - R_D I_{SS}/2$

➔ $V_X - V_Y = 0$



That is, a zero differential input (i.e Common mode signal) gives a zero differential output.

Output CM level is equal to $V_{DD} - R_D I_{SS}/2$.

2

Differential amplifier

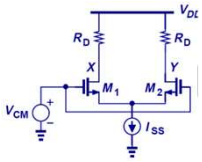
Qualitative analysis (contd..):

Equilibrium Overdrive Voltages of M_1, and M_2 , $(V_{GS}-V_{TH})_{equil.}$:

Assume $\lambda = 0$, hence $I_D = (1/2)\mu_n C_{ox}(W/L)(V_{GS} - V_{TH})^2$ carrying a current of $I_{SS}/2$, each device exhibits an overdrive of

$$I_D = \frac{I_{SS}}{2} = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS}-V_{TH})^2$$

$$\Rightarrow (V_{GS}-V_{TH})_{equil} = \sqrt{\frac{I_{SS}}{\mu_n C_{ox}\frac{W}{L}}}$$



The equilibrium overdrive voltage is defined as the overdrive voltage seen by M1 and M2 when both of them carry a current of ISS/2.

As expected, a greater tail current or a smaller W/L translates to a larger equilibrium overdrive.

3

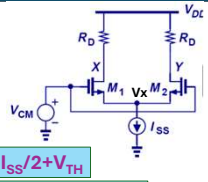
Differential amplifier

Minimum Common-mode Voltage:

To guarantee that M1 and M2 operate in saturation, we require that their drain voltages not fall below $V_{CM} - V_{TH}$:

Condition for saturation: $V_D \geq V_G - V_{TH}$

Here $V_G = V_{CM}$ and VD1=VD2= $V_{DD} - R_D I_{SS}/2$



$$V_{DD} - R_D\frac{I_{SS}}{2} > V_{CM} - V_{TH}$$

$V_{CM} \leq V_{DD} - R_D I_{SS}/2 + V_{TH}$

$V_{CM}\,(max) = V_{DD} - R_D I_{SS}/2 + V_{TH}$

If the values is greater than above, then MOSFET enters into triode region

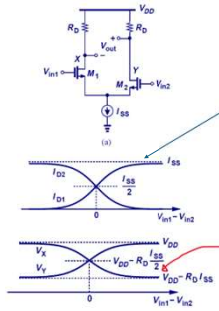
Minimum value	Maximum value
$(-V_{ss} + V_X + V_{GS}) \leq V_{CM} \leq [V_{DD} - (I_{ss}/2)\ R_D + V_{TH}]$	

Range of CM signal for proper DA operation

4

Differential amplifier

Differential Response:



Response of the MOS pair to large differential inputs:

Case1:

- **If Vin1 > Vin2 [Fig. a.], then**
- **ID1>ID2 because more current flows through M1**
- **Therefore, more drop across RD of M1 ➔ VX ↓ ↓**
- But w.k.t : ID1+ID2=ISS
- As a result, Less drop across RD of M2 ➔ VY ↑ ↑
- So, as Vin1-Vin2 becomes more ++ve, VX-VY becomes more –ve.
- At one point, M1 draws entire tail current making ID2=0 (M2 off) and ID1=ISS. At the point,

$V_X = V_{DD} - R_D I_{SS}$

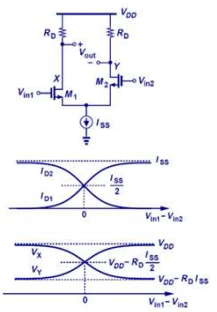
$V_Y = V_{DD}$

$V_X - V_Y = -I_{SS}R_D$

5

Differential amplifier

Differential Response:



Response of the MOS pair to large differential inputs:

Case2:

- **Similarly, if Vin2 > Vin1, then**

$V_X = V_{DD}$

$V_Y = V_{DD} - R_D I_{SS}$

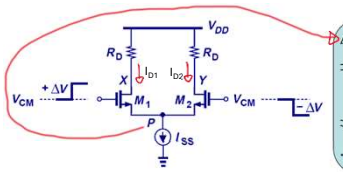
$V_X - V_Y = +I_{SS}R_D$

- **The circuit therefore steers the tail current from one side to the other, producing a differential output in response to a differential input.**

6

Differential amplifier

Response of MOS pair to small differential inputs:



$$\Delta V_P = 0$$
$$\Rightarrow \Delta I_{D1} = g_m \Delta V,$$
$$\Delta I_{D2} = -g_m \Delta V$$
$$\Rightarrow \Delta V_X - \Delta V_Y = -2 g_m R_D \Delta V$$
$$\therefore A_v = -g_m R_D$$

$$g_m = \frac{\partial I_D}{\partial V_{GS}}$$

The obtained differential voltage gain is similar to that of a common-source stage.

7

MOS Differential Pair's Large-Signal Analysis:

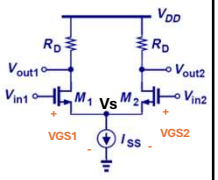
Objective here is to derive the input/output characteristics of the MOS pair as the differential input varies from very negative to very positive values.

Consider ID1 =ISS; ID2=0

Vin1-VGS1=VS

Vin2-VGS2=VS

Goal is to obtain $I_{D1} - I_{D2}$



$$V_{in1} - V_{GS1} = V_{in2} - V_{GS2}.$$
$$I_{D1} + I_{D2} = I_{SS}.$$
$$I_D = (1/2)\mu_n C_{ox}(W/L)(V_{GS} - V_{TH})^2.$$
$$\Rightarrow V_{GS} = V_{TH} + \sqrt{\frac{2I_D}{\mu_n C_{ox}\frac{W}{L}}}.$$
$$\Rightarrow V_{in1} - V_{in2} = V_{GS1} - V_{GS2}$$
$$= \sqrt{\frac{2}{\mu_n C_{ox}\frac{W}{L}}}\left(\sqrt{I_{D1}} - \sqrt{I_{D2}}\right).$$

8

Differential amplifier

MOS Differential Pair's Large-Signal Analysis (contd..):

$$(V_{in1} - V_{in2})^2 = \frac{2}{\mu_n C_{ox}\frac{W}{L}}(I_{D1} + I_{D2} - 2\sqrt{I_{D1} I_{D2}})$$
$$= \frac{2}{\mu_n C_{ox}\frac{W}{L}}(I_{SS} - 2\sqrt{I_{D1} I_{D2}}).$$
$$\Rightarrow 4\sqrt{I_{D1} I_{D2}} = 2I_{SS} - \mu_n C_{ox}\frac{W}{L}(V_{in1} - V_{in2})^2$$
$$\Rightarrow 16 I_{D1} I_{D2} = \left[2I_{SS} - \mu_n C_{ox}\frac{W}{L}(V_{in1} - V_{in2})^2\right]^2$$
$$\Rightarrow 16 I_{D1}(I_{SS} - I_{D1}) = \left[2I_{SS} - \mu_n C_{ox}\frac{W}{L}(V_{in1} - V_{in2})^2\right]^2$$

$$\Rightarrow 16 I_{D1}^2 - 16 I_{SS} I_{D1} + \left[2I_{SS} - \mu_n C_{ox}\frac{W}{L}(V_{in1} - V_{in2})^2\right]^2 = 0$$
$$\Rightarrow I_{D1} = \frac{I_{SS}}{2} \pm \frac{1}{4}\sqrt{4I_{SS}^2 - \left[\mu_n C_{ox}\frac{W}{L}(V_{in1} - V_{in2})^2 - 2I_{SS}\right]^2}$$
$$= \frac{I_{SS}}{2} + \frac{V_{in1} - V_{in2}}{4}\sqrt{\mu_n C_{ox}\frac{W}{L}\left[4I_{SS} - \mu_n C_{ox}\frac{W}{L}(V_{in1} - V_{in2})^2\right]}$$
$$\Rightarrow I_{D2} = \frac{I_{SS}}{2} + \frac{V_{in2} - V_{in1}}{4}\sqrt{\mu_n C_{ox}\frac{W}{L}\left[4I_{SS} - \mu_n C_{ox}\frac{W}{L}(V_{in2} - V_{in1})^2\right]}$$

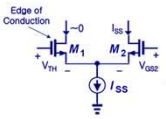
($\because$ the symmetry of the circuit)

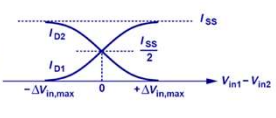
$$I_{D1} - I_{D2} = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{in1} - V_{in2})\sqrt{\frac{4I_{SS}}{\mu_n C_{ox}\frac{W}{L}} - (V_{in1} - V_{in2})^2}$$

9

Maximum Differential Input Voltage:

- As |Vin1 – Vin2| rises, at some point M1 or M2 turns off, violating the above equation.
- We must therefore determine the input difference that places one of the transistors at the edge of conduction.





- For this consider, M1 approaches the edge of conduction

➔ $V_{GS1} = V_{TH}$

$$V_{GS2} = V_{TH} + \sqrt{\frac{2I_{SS}}{\mu_n C_{ox}\frac{W}{L}}}.$$

The input difference that places one transistor at the edge of conduction

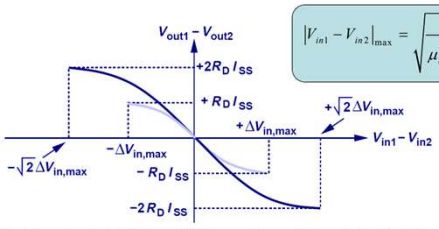
$$|V_{in1} - V_{in2}|_{max} = \sqrt{\frac{2I_{SS}}{\mu_n C_{ox}\frac{W}{L}}} = \sqrt{2}(V_{GS} - V_{TH})_{equil}$$

There exists a finite differential input voltage that completely steers the tail current from one transistor to the other. This value is known as the maximum differential input voltage.

10

Differential amplifier

The effects of Doubling the Tail Current:



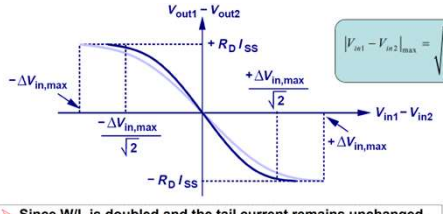
$$|V_{in1} - V_{in2}|_{max} = \sqrt{\frac{2I_{SS}}{\mu_n C_{ox}\frac{W}{L}}} = \sqrt{2}(V_{GS} - V_{TH})_{equil}$$

- Since I_{SS} is doubled and W/L is unchanged, the equilibrium overdrive voltage for each transistor must increase by $\sqrt{2}$ to accommodate this change, thus $\Delta V_{in,max}$ increases by $\sqrt{2}$ as well. Moreover, since I_{SS} is doubled, the differential output swing will double.

11

Differential amplifier

The effects of Doubling W/L:



$$|V_{in1} - V_{in2}|_{max} = \sqrt{\frac{2I_{SS}}{\mu_n C_{ox}\frac{W}{L}}} = \sqrt{2}(V_{GS} - V_{TH})_{equil}$$

- Since W/L is doubled and the tail current remains unchanged, the equilibrium overdrive voltage will be lowered by $\sqrt{2}$ to accommodate this change, thus $\Delta V_{in,max}$ will be lowered by as well. Moreover, the differential output swing will remain unchanged since neither I_{SS} nor R_D has changed

12

Differential amplifier

COMMON-MODE REJECTION RATIO (CMRR):

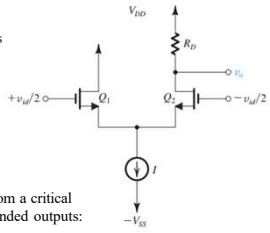
CMRR defines the ratio of wanted amplified differential input signal to unwanted converted input common-mode noise that appears at the output

$$CMRR = \frac{A_{DM}}{A_{CM}}$$

13

Differential-to-Single-Ended Conversion

- Ignore the drain current signal of $Q1$ and eliminate its drain resistor.
- Take the output between the drain of $Q2$.
- Drawback of this scheme is that we lose a factor of 2 (or 6 dB) in gain as a result of "wasting" the drain signal current of Q1.



Problem with Simple Differential Pairs

Traditional differential pairs with resistive loads suffer from a critical limitation when converting differential signals to single-ended outputs:

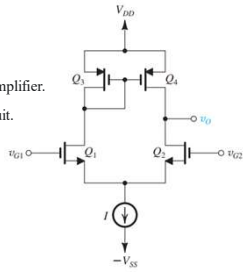
- Only one transistor's output is utilized
- The other transistor's drain signal current is "wasted"
- This Results in a 6 dB gain loss (factor of 2 reduction)
- → Inefficient use of available signal current

This inefficiency led to the development of current-mirror-loaded topology.

14

The Current-Mirror-Loaded MOS Differential Pair

- **Q_1 and Q_2** form the differential amplifier pair.
- **Q_3 and Q_4** form a current mirror that acts as the load for the amplifier.
- **The current source I sets the bias current** for the entire circuit.



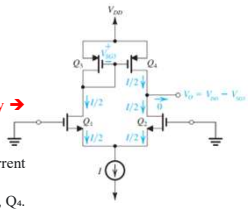
15

Principle of Operation

1. Common mode operation

Imagine we apply the same DC voltage to both inputs ($v_{id} = 0$)

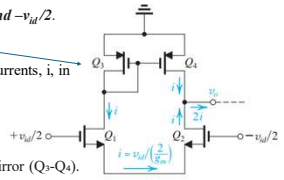
- **Current Splitting:**
  - Assuming perfect matching, the bias current I splits equally → the drain currents of Q_1 and Q_2 = I/2.
- **Current Mirror Action:**
  - The drain current of Q_1 (I/2) is fed into the input of the current mirror.
  - The mirror then "copies" this current to its output transistor, Q_4. Therefore, the drain current of Q_4 is also I/2.
- **Output Node:**
  - At the output node (v_o), we have two currents: I/2 flowing down through Q_2 and I/2 flowing down into Q_4.
  - These currents perfectly balance each other, leaving zero current to flow to the next stage.
  - If Q4 is perfectly matched to Q3, its drain voltage will track the voltage at the drain of Q3
    - → $V0 = V_{DD} - V_{SG3}$ …….. (But how we got this equation??)



16

2. Differential mode operation

- Now, **let's apply differential input signal, $+v_{id}/2$ and $-v_{id}/2$.**
- Small-signal circuit will look like:
- The input ($v_{id}/2$) creates equal and opposite signal currents, i, in the drains of Q_1 and Q_2.

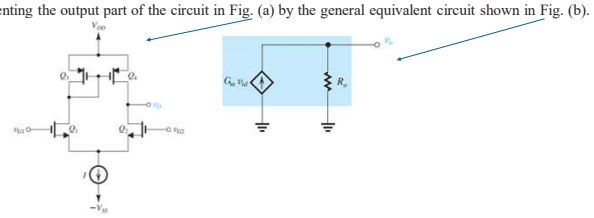


- The signal current i is given by $i = gm * (v_{id}/2)$
- The signal current i from Q_1 flows into the current mirror (Q_3-Q_4).
- The mirror responds by creating a replica of this current in the drain of Q_4.
- This means a signal current i is now drawn out of the output node by Q_4.
- At the output node, the two signal currents now add together.
- The combine to form a total output current of 2i that flows to the load
- **Key to the circuit's effectiveness:** for differential signals, the current mirror adds its output current to the current of Q_2, doubling the output current and converting the signal to a single-ended form with no loss of gain.

17

Differential Gain of the Current-Mirror-Loaded MOS Pair

Representing the output part of the circuit in Fig. (a) by the general equivalent circuit shown in Fig. (b).



- Here *Gm* is the short-circuit transconductance and *Ro* is the output resistance
- $G_m = g_{m1,2}$
  - Here $g_{m1,2}$ is the transconductance of each of Q1 and Q2.
- $R_o = ro2 \parallel ro4$

18

$$A_d \equiv \frac{v_o}{v_{id}} = G_m R_o = g_{m1,2}(r_{o2} \| r_{o4})$$

Writing $g_{m1,2}$ simply as g_m, and for the case $r_{o2} = r_{o4} = r_o$,

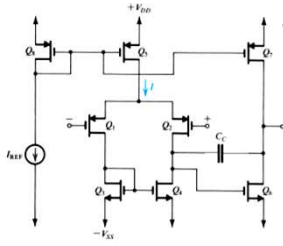
$$A_d = \frac{1}{2} g_m r_o = \frac{1}{2} A_0$$

where A_0 is the intrinsic gain of the MOS transistor.

19

Differential amplifier

Analysis of two stage CMOS Operational amplifier with single ended output:



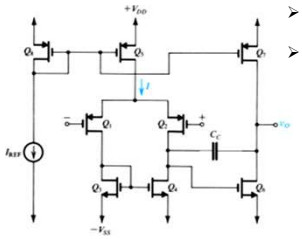
The circuit consists of **two gain stages**:

- **First stage:** Differential pair **Q1– Q2** together with its current-mirror load **Q3–Q4**.
- The differential pair is biased by current source Q5, which is one of the two output transistors of the current mirror formed by Q8, Q5, and Q7.
- The current mirror is fed by a reference current I_{REF}, which can be generated by simply connecting a resistor (external to the chip) to the negative supply voltage – VSS or to a more precise negative voltage reference if one is available in the same integrated circuit.

20

Differential amplifier

Analysis of two stage CMOS Operational amplifier with single ended output:



- **Second gain stage:** Consists of the common-source transistor *Q*6 and its current-source.
- It takes part in the process of frequency compensating the op amp.

21

Voltage Gain

- The voltage gain of the first stage $A_1 = -g_{m1}(r_{o2} \| r_{o4})$

  where g_{m1} is the transconductance of each of the transistors of the first stage, that is, *Q*1 and *Q*2.

- The second stage is a current-source-loaded, common-source amplifier whose voltage gain is given by $A_2 = -g_{m6}(r_{o6} \| r_{o7})$

- The dc open-loop gain of the op amp $Av = A1 * A2$

- **A striking feature:** It does *not* have a low-output-resistance stage.
- Output resistance Zout = (*ro* ||*ro*7) and is thus rather high.
- This circuit, is not suitable for driving low-impedance loads.
- Nevertheless, the circuit is very popular and is used frequently for implementing op amps in VLSI circuits, where the op amp needs to drive only a small capacitive load

22

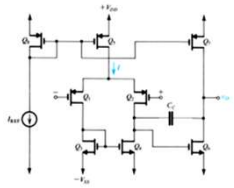
Input Offset Voltage

- The device mismatches inevitably present in the input stage give rise to an input offset voltage.
- Unless properly designed, due to mismatches, the CMOS op-amp circuit can exhibit a **systematic output dc offset** voltage.
- The systematic dc offset can be eliminated by sizing the transistors so as to satisfy the following constraint:

$$\frac{(W/L)_6}{(W/L)_4} = 2\frac{(W/L)_7}{(W/L)_5}$$

Note:

$$I_6 = \frac{(W/L)_6}{(W/L)_4}(I/2) \qquad I_7 = \frac{(W/L)_7}{(W/L)_5} I$$



23